

Full Envelope Airspeed Calibration



Introduction

- What is static source position error (SSPE)?
- Kinematic relationships for airspeed calibration
- Traditional methods for airspeed calibration
- The single turning acceleration technique
- Numerical results from C-130J simulations
- Real flight test results



Static Source Position Error (SSPE)

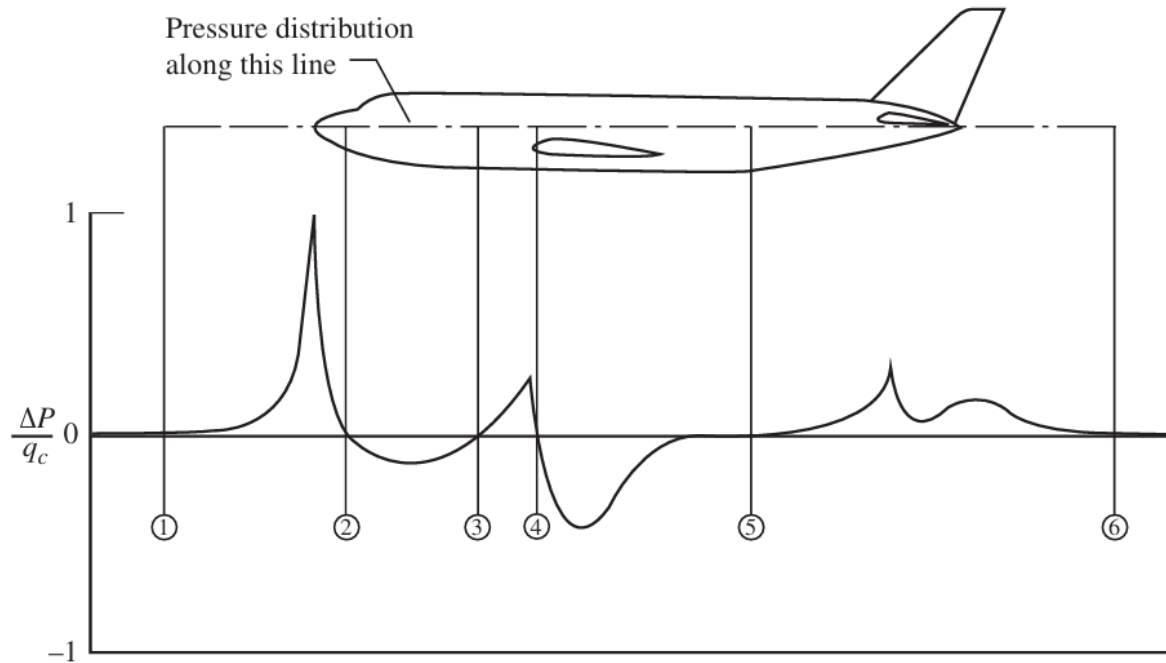


Figure 1 The pitot-static probe mounted on an airplane and the surface pressure distribution along the fuselage reference line (FRL). Source: Haering (1995).

- $V_{CAS} = V_{IAS} + \Delta V$
- SSPE is caused by aircraft-induced static pressure distortion
- Airspeed errors are also functions of airspeed (and by corollary, a function of Angle of Attack (AOA) and load factor).



Kinematic Relationships for Airspeed Calibration

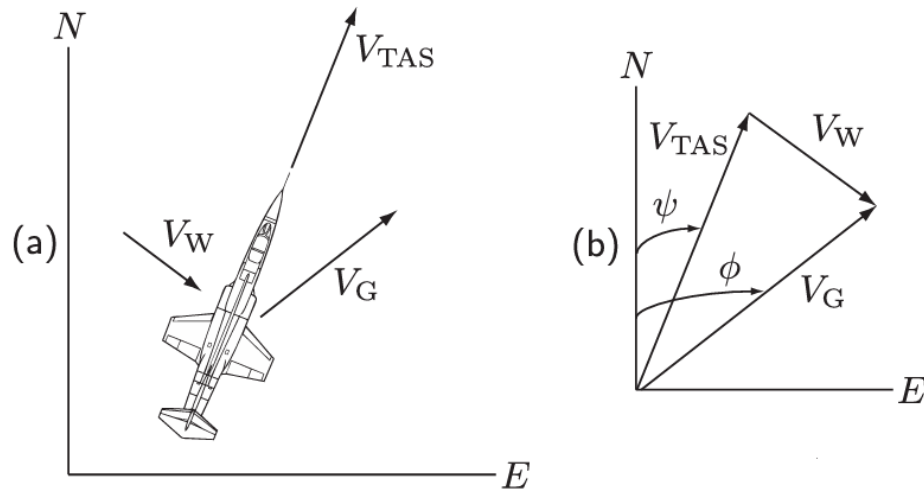


Figure 2: Reference System Definition.

IAS and TAS direction → **heading ψ**

Ground speed direction → **course ϕ**

Wind vector → **unknown**



Kinematic Relationships for Airspeed Calibration

$$V_{CAS} = a_0 \sqrt{5 \left[\left(\frac{q_c}{P_0} + 1 \right)^{\frac{2}{7}} - 1 \right]}$$
$$V_{TAS} = a_0 \sqrt{5 \frac{T}{T_0} \left[\left(\frac{q_c}{P} + 1 \right)^{\frac{2}{7}} - 1 \right]}$$

$$V_{CAS} = V_{IAS} + \Delta V$$

Assume sea-level flight:

$$T = T_0, \quad P = P_0$$

$$V_{CAS} = V_{TAS}$$

$$V_{TAS} + V_w = V_G$$

$$\boxed{V_{IAS} + \Delta V + V_w = V_G}$$



GPS-Based Cloverleaf Calibration Method

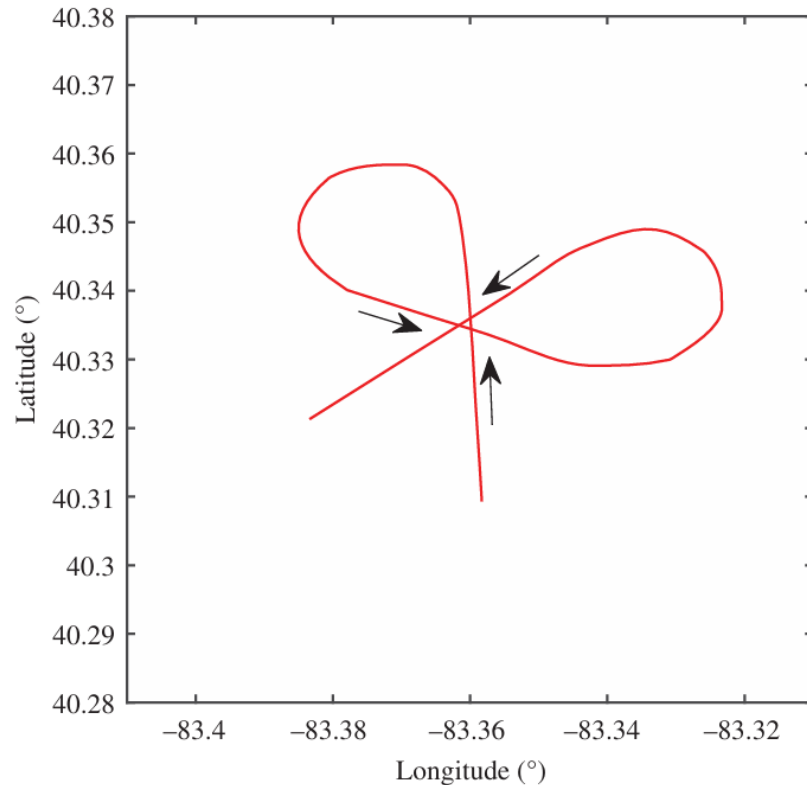


Figure 3 Sample ground track (a “cloverleaf”) for GPS-based determination of TAS.

Three Unknowns in the Cloverleaf Method:

ΔV

$V_{wind,E}$

$V_{wind,N}$

$$V_{IAS} + \Delta V = V_{TAS} = \sqrt{(V_{GS,E} - V_{wind,E})^2 + (V_{GS,N} - V_{wind,N})^2}$$



Level Turn Airspeed Calibration Method

North Component

$$V_G \cos \phi = (V_{IAS} + \Delta V_n) \cos \psi + V_{W_N}$$

East Component

$$V_G \sin \phi = (V_{IAS} + \Delta V_n) \sin \psi + V_{W_E}$$

The Matrix Form for Single Point

$$\begin{bmatrix} V_G \cos \phi - V_{IAS} \cos \psi \\ V_G \sin \phi - V_{IAS} \sin \psi \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cos \psi \\ 0 & 1 & \sin \psi \end{bmatrix} \begin{bmatrix} V_{W_N} \\ V_{W_E} \\ \Delta V_n \end{bmatrix}$$

For $M = 10^3$ Points

$$\begin{bmatrix} V_{G(1)} \cos \phi_{(1)} - V_{IAS(1)} \cos \psi_{(1)} \\ \vdots \\ V_{G(M)} \cos \phi_{(M)} - V_{IAS(M)} \cos \psi_{(M)} \\ V_{G(1)} \sin \phi_{(1)} - V_{IAS(1)} \sin \psi_{(1)} \\ \vdots \\ V_{G(M)} \sin \phi_{(M)} - V_{IAS(M)} \sin \psi_{(M)} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cos \psi_{(1)} \\ \vdots & \vdots & \vdots \\ 1 & 0 & \cos \psi_{(M)} \\ 0 & 1 & \sin \psi_{(1)} \\ \vdots & \vdots & \vdots \\ 0 & 1 & \sin \psi_{(M)} \end{bmatrix} \begin{bmatrix} V_{W_N} \\ V_{W_E} \\ \Delta V_n \end{bmatrix}$$

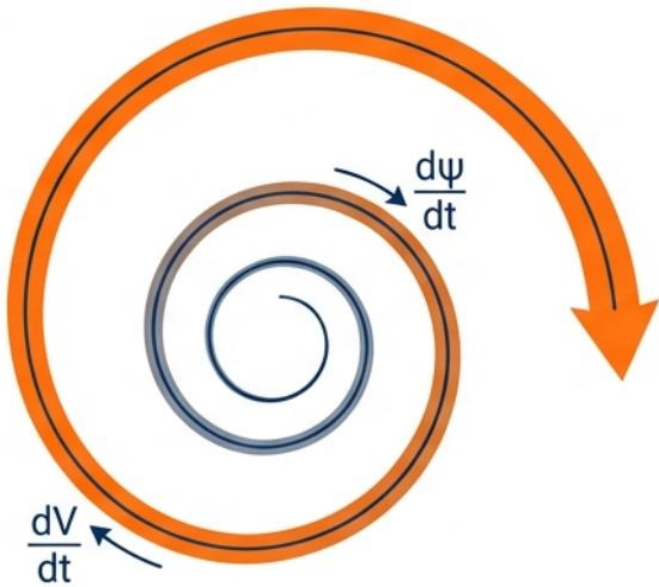
- **Overdetermined system**

- No unique solution → **least squares**

- **Wind magnitude and direction are inferred.**



The Single Turning Acceleration Technique



- Airspeed is intentionally accelerated during the turn

$$dV/dt \neq 0$$

- Heading changes continuously

$$d\psi/dt \neq 0$$

- Each time sample provides an **independent observation** of the wind and airspeed error relationship
- The **static source position error** is modeled as a **function of IAS** rather than a constant bias



The Single Turning Acceleration Technique

Model Name	Mathematical Form	Type
Polynomial(N)	$\mathcal{M}_{\Delta}(V_{IAS}; \mathbf{w}) = \sum_{i=0}^P w_i V_{IAS}^i$	Linear
Power	$\mathcal{M}_{\Delta}(V_{IAS}; \mathbf{w}) = w_0 + w_1 V_{IAS}^{w_2}$	Nonlinear
Exponential	$\mathcal{M}_{\Delta}(V_{IAS}; \mathbf{w}) = w_0 + w_1 \exp(w_2 V_{IAS})$	Nonlinear

Table 1: Example model families used in fitting SSPE curves.

Several **candidate error model families** are considered:

- Polynomial (flexible, linear in parameters)
- Power-law (smooth monotonic behavior)
- Exponential (rapid nonlinear trends)
- Akaike Information Criterion (AIC/AICc) balances goodness of fit against model complexity.



The Single Turning Acceleration Technique

Altitude Selection:

- As close as possible to **sea level** or at **higher altitudes** with additional computational and instrumentation requirements.
- The maneuver tolerance should be **± 50 feet** to assure a confidence interval of **± 0.5 knots**.

Angle of Bank Selection:

- Angle of bank directly determines **turn radius** and **load factor**.
- Increasing AOB increases **load factor**, which in turn increases **angle of attack (AoA)** for a given airspeed.
- Therefore, the **lowest practical angle of bank** is recommended to best approximate wings-level airspeed calibration



The Single Turning Acceleration Technique

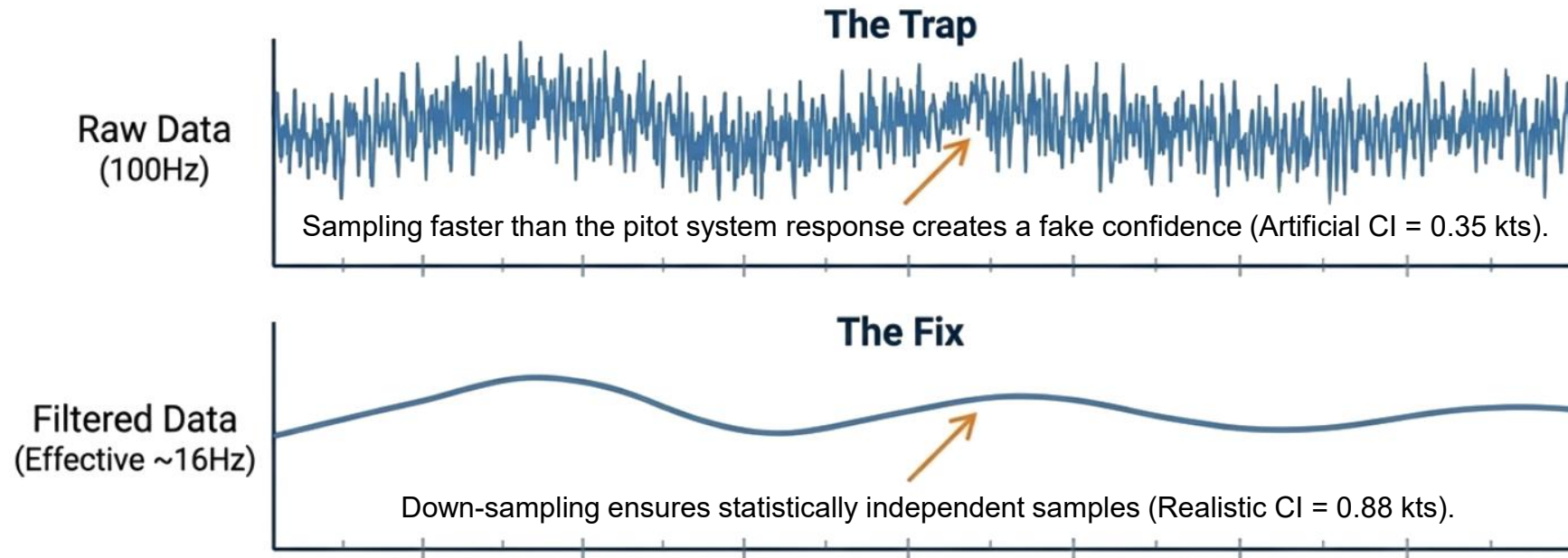
Acceleration Profile:

- At least **one full turn** is required to resolve **wind magnitude and direction**
- A **wide range of airspeeds** is necessary to accurately estimate the SSPE curve
- The acceleration profile should remain **as constant as possible** to minimize wind variability effects
- Separate tests are recommended for **different airspeed regimes** (e.g. subsonic vs transonic)



The Single Turning Acceleration Technique

Instrumentation Data Rate



- More data is **not** always better. Sequential correlation in high-speed sensors can artificially shrink confidence intervals.

(SFTE)



Numerical Results from C-130J Simulations

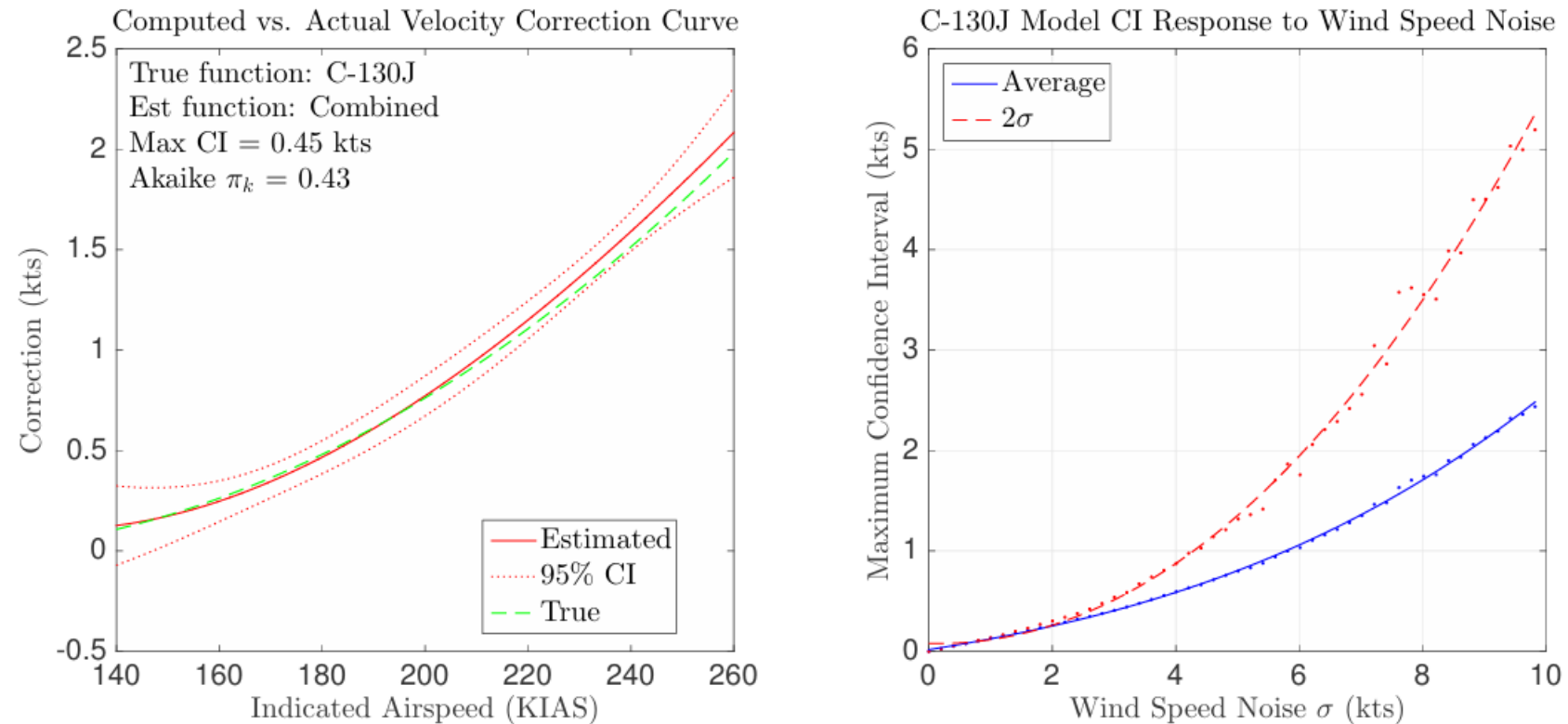


Figure 4: C-130J Numerical Results



Real Flight Test Results

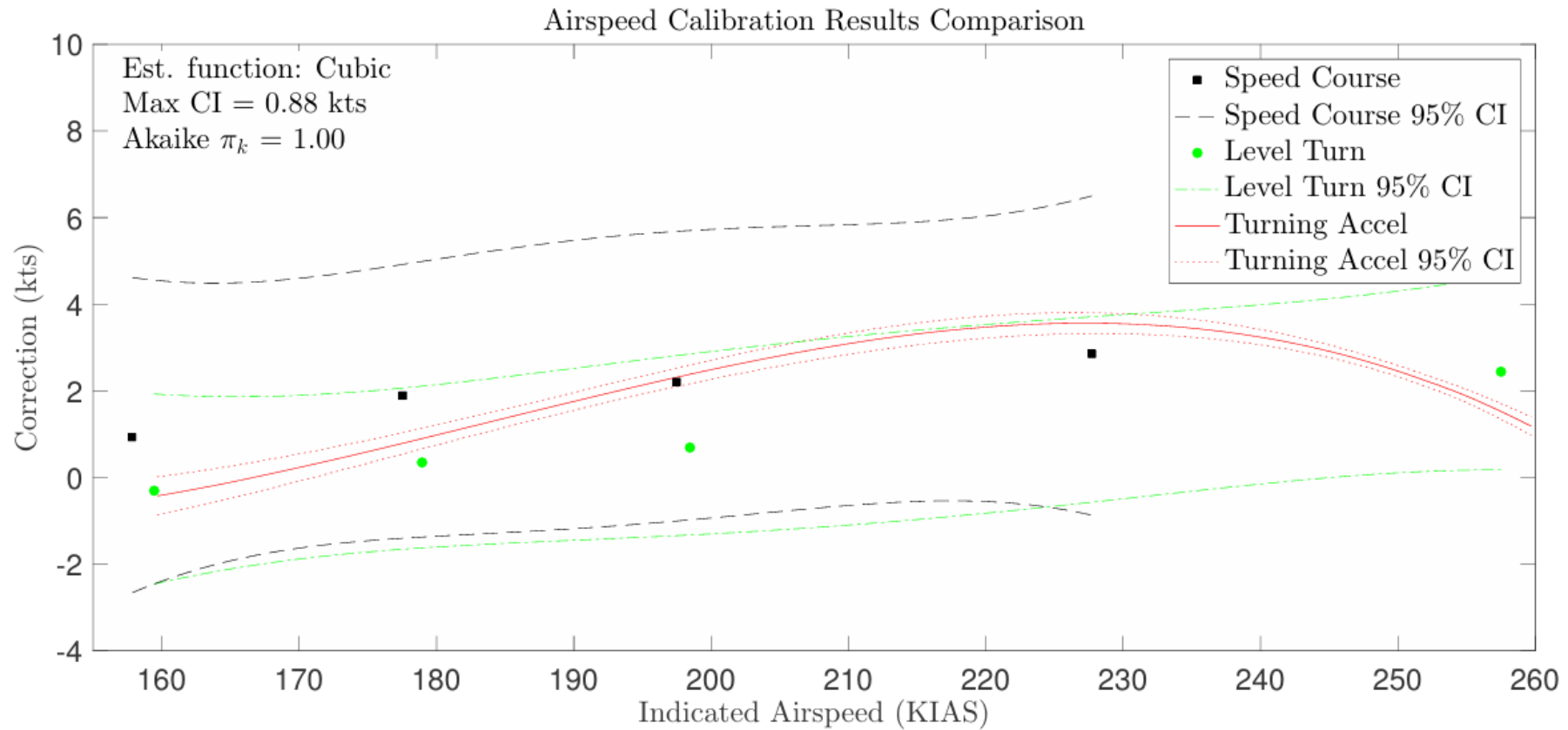


Figure 5: Flight Test Results.



Real Flight Test Results

	FTT	Test Points	Flight Time	Savings
Ground Speed Course		4	36.5 min.	-
Level Turn		4	15.7 min.	57%
Turning Acceleration		1	4.6 min.	87%

Table 2: Flight test time comparison.

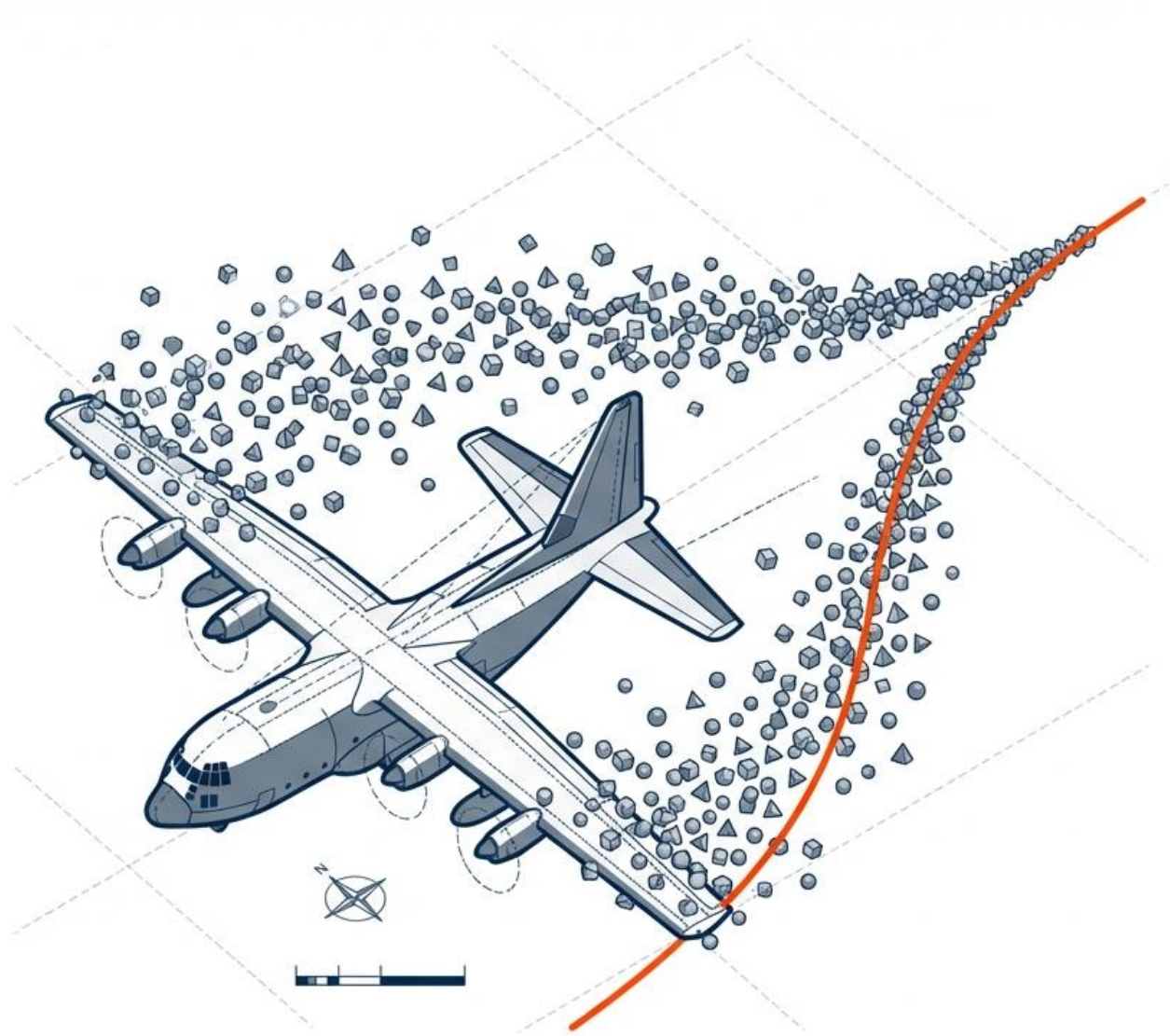
In **level turn methods**, gust-affected data points are relatively easy to identify.

- Measurements showing sudden deviations are **discarded** from the analysis
- The results are more accurate for a single **IAS value** (Assuming no drift in wind).
- But flying the whole envelope results in **higher** costs.
- Less tolerant of execution error due to longer flight times.

In the **turning acceleration technique**, airspeed is intentionally varied.

- Gust effects are treated as **disturbances**.
- Their influence is **reduced numerically** through regression and averaging
- Data are **down-weighted**, not discarded
- Because turning acceleration method is quick to execute, it tends to be less affected by wind drifts (not gusts).





Thank you.



References

- [1] J. D. Jurado, C. C. McGehee, and T. R. Jorris, "Full envelope airspeed calibration using a single turning acceleration technique," in *Proceedings of the 47th Annual International Symposium of the Society of Flight Test Engineers (SFTE)*, Wichita, KS, USA, Aug. 2016.
- [2] I. M. Gregory and Y. Liu, *Introduction to Flight Testing*. Wiley, 2021.
- [3] R. E. Erb, *Pitot-Statics Textbook*. Edwards AFB, CA, USA: USAF Test Pilot School, 2010.
- [4] T. R. Jorris, M. M. Ramos, R. E. Erb, and R. K. Woolf, "Statistical pitot-static calibration technique using turns and self-survey method," in *Proceedings of the 42nd International SFTE Symposium*, Seattle, WA, USA, 2011.
- [5] H. Akaike, "A new look at the statistical model identification," *IEEE Transactions on Automatic Control*, vol. 19, no. 6, pp. 716–723, Dec. 1974.

